

# **Institute of Actuaries of India**

## **Subject ST7 – General Insurance: Reserving & Capital Modelling**

### **October 2014 Examination**

#### **INDICATIVE SOLUTION**

**Solution 1:** *All figures below are in INR lacs*

- i) Risk attaching basis covers all policies written or renewed on or after the treaty date, and before expiration. This is claim # 4, 5, 6, 7, 8

$$\text{Total losses} = 65 + 120 + 400 + 185 + 90 = 860 \quad [2]$$

- ii) Losses occurring basis covers all policies written or renewed after the treaty eff date + the unearned portion of policies in-force as of the treaty date.

This is claim # 1, 3, 4, 5, 6, 7, 8

$$\text{Total loss} = (\text{loss from part A} + 50 + 220) = 1130 \quad [2]$$

- iii) In-force policy basis covers only the unearned portion of policies inforce as of the treaty effective date. This is claim # 1 & 3

$$\text{Total loss} = 50 + 220 = 270 \quad [2]$$

[6 Marks]

**Solution 2:**

Given solution is just one of the various approaches to solve this problem. Other reasonable approaches must also be given credit. All figures in INR crores

- i) Ceding Company:

$$\text{Assets} = 180 + 0.25(80) - 80 = 120$$

$$\text{UEPR} = 120 - 80 = 40$$

$$\text{P/H Surplus} = 60 - 0.25(80) = 80$$

$$\text{Reinsurer} = 300 + 80 - 0.25(80) = 360$$

$$\text{UEPR} = 220 + 80 = 300$$

$$\text{P/H Surplus} = 60 \quad [3]$$

- ii) 1) They may be ceding away profitable business (in exchange for Surplus Relief)

2) If commission on the reinsurance is also proportional, it may not cover the expenses of the cedant

[1]

[4 Marks]

**Solution 3: Reporting delays**

- Might arise because, say, an injury gets worse over time whereas the sufferer had expected the problem to be only temporary, or because the seriousness of the condition only comes to light in a later routine check-up.
- Some liability claims remain undetected for some years. This means that loss may not become evident until many years after it was caused. This exacerbates the long-tailed nature of the associated liability insurance if it was written on a losses-occurring basis.
- A further complication is that claims may arise over a long period; for example a storage tank may leak contents into the ground, causing pollution, for some years before the damage is discovered. This means that it may not be clear which insurance should respond to the loss, and often some form of allocation is necessary.

**Settlement delays**

- Settlement delays may be due to the delays caused by establishing liability, or the time to establish the extent of injuries and to assess the speed of recovery.
- If the loss is personal injury, it may take some years before the victim's condition has stabilized and can be assessed for damages.
- Claim settlement will involve litigation for any claim of significant size, even though most cases are resolved out of court before they go to trial.
- Settlement for motor liability is often faster than for other types of liability claims, although claims arising from serious bodily injury do typically take several years to settle. Even so, motor liability claims, and particularly bodily injury claims, are significantly longer tailed than other motor claims.
- The legal environment has a significant effect on the settlement of claims, and this varies from country to country. Therefore claims development patterns for similar classes of business may vary significantly between countries.

**[6 Marks]**

**Solution 4:**

Fire and weather related perils: Protection from damage to restaurant building, fixtures and contents due to fire, weather and the like.

Product liability: Protection against claims related to products sold such as food poisoning claims or your fast food is blamed for making a customer sick.

Premises liability: Protection against occurrences on the restaurant premises such as slip-and-fall injuries or accidentally injured while dining at your restaurant

Business Interruption: Protection against loss of income if the business has to close temporarily due to fire or deterioration of stocks or food spoilage etc.

Workers Compensation: Protection against liability under the Workmen to pay compensation to employees for bodily injury or death caused due to accidents / occupational diseases arising out of and in course of employment

Motor Fleet: Motor Insurance to protect the vehicles owned (two wheelers for the purpose of home delivery, as an example) against any damage due to accidents and also against any third party liability due to accidents. Motor Third Party insurance is also a mandatory requirement.

Other needs: financial loss due to fraud or dishonesty of the employee, loss due to Burglary or Housebreak; losses due to deterioration of stocks caused by an unexpected rise in temperature brought about by the accidental failure or breakdown of the refrigeration plant; unforeseen and sudden physical damage due to electrical or mechanical breakdown e.g. air-conditioners, refrigeration etc.; damage to plate glass or neon signs; marine transit cover any goods/stock being transferred to the locations. May require to provide group health insurance and/or group personal accident insurance to its employees

**[8 Marks]**

**Solution 5:**

Other reasonable solutions must be given due credit

i.) Reported = Cumulative Paid + Cumulative Outstanding =  $310 + 45 = 355$  **[2]**

ii) Paid claims in 2013 =  $85 + (238 - 115) + (310 - 165) + (337 - 288) = 402$  **[2]**

iii) 2012 case outstanding =  $115 + 94 + 74 = 283$   
 2013 case outstanding =  $208 + 238 + 45 + 35 = 526$ .  
 Change in case reserves =  $526 - 283 = 243$ . **[2]**

iv) Numerous benchmarks are tracked by accident year.  
 Tracking claims by accident year can be very useful when economic/regulatory changes have recently occurred, or if a significant large loss has occurred.  
 Exposures corresponding to a given accident year were subject to the same risk environment.  
 Translates easily into the accounting requirements. **[2]**

v) The effect of the unadjusted reported loss method will be ultimate loss overstatement. The method will calculate development factors based on historical emergence and developments and apply them to the latest

reported values to determine ultimate estimates. Since case and paid will increase for the latest year (since case reserves are higher and payment rate will increase with settlement), the projected losses will be higher when the larger LDF is applied to larger emerged values.

[2]

vi) Similar to part a., paid loss development will be overstated. The change in case will not cause any changes but the increase in settlement will cause the distortion. More will be paid earlier as claims settle faster. Once again, the higher LDF application will cause overstatement.

- vii) 1) Has the underwriting department lowered standards in regards to what constitutes an acceptable insured?  
 2) Any changes made to Underwriting terms and conditions?  
 3) Any changes to the Reinsurance Treaty terms?  
 4) Has there been a change in geographic area / business mix regarding new underwriting strategies?

[2]

[14 Marks]

**Solution 6:**

i)

| Cum Paid Losses |        |        |        |        |
|-----------------|--------|--------|--------|--------|
| AY              | 12     | 24     | 36     | 48     |
| 09-10           | 27,000 | 49,000 | 65,000 | 72,000 |
| 10-11           | 28,000 | 57,000 | 71,000 | 81,650 |
| 11-12           | 33,000 | 65,000 | 78,000 | 89,700 |
| 12-13           | 35,000 | 70,000 | 84,000 | 96,600 |

| Incremental Future Paid Losses |        |        |        |
|--------------------------------|--------|--------|--------|
| AY                             | 12-24  | 24-36  | 36-48  |
| 09-10                          |        |        |        |
| 10-11                          |        |        | 10,650 |
| 11-12                          |        | 13,000 | 11,700 |
| 12-13                          | 35,000 | 14,000 | 12,600 |

$$\text{Undiscounted} = 35,000 + 13,000 + 10,650 + 14,000 + 11,700 + 12,600 \\ = 96,950$$

$$\text{Discounted} = \\ (35,000 + 13,000 + 10,650)/1.06^{.5} + (14,000 + 11,700)/1.06^{1.5} + (12,600)/1.06^{2.5} \\ = 56,966 + 23,549 + 10,892 \\ = 91,407$$

Discounted/Undiscounted= $91407/96,950=0.9428$  [3]

- ii) When the policy limits or deductible profile of a book of business has changed from the previous. Policy year should be used instead of AY.  
Or if there were any changes made to the Reinsurance Treaty terms.

[2]

- iii) Accident quarter should be used instead of AY when the exposures of the book are growing rapidly resulting in a shift of the average accident date.  
If there is any seasonality observed in claims experience.

[2]

- iv) Report year data should be used instead of AY data when a drastic change in the social or legal climate causes average severity to more closely correlate with the date of the report of the claim.

For claims-made policies, analysis of cohorts by RY instead of AY is useful as only the development of reported claims is to be studied.

For analysis of claim settlement delay patterns.

For estimation of IBNER.

[2]

- v) Earned exposures should be used instead of claim counts when the definition of claim counts has changed and would cause a distortion if used.

For long-tailed lines where the claim counts are low as they are yet to develop. Or for low-frequency, high-severity lines.

Reserving for catastrophe events in the initial stages when only the exposure to the event is known.

[2]

[11 Marks]

### **Solution 7:**

i)

Many employers' liability claims are reported and settled quickly.

However, some may take some time to determine / agree liability, some of which may go to court and experience lengthy delays.

Also the potential for latent claims is considerable.

Settlements are likely to be in real terms.

Marine and Aviation claims may be for damage and/or liability.

Delays in reporting are not significant.

However, settlement delays, especially for liability claims can often be lengthy due to the number of parties involved.

Private motor claims are mostly reported and settled quickly.

A few liability claims may take a few years to settle.  
 Many claims are for repairs and will be affected by wage, inflation and the level of prices of motor vehicle parts.  
 Country of business and hence currency of liabilities

[4]

ii)

If used it is likely to reflect the total return on assets backing the technical reserves.  
 May use a different discount rate for different parts of the business.  
 Where mismatched a notional portfolio reflecting a more matched position may be used instead.  
 A lower discount rate than that calculated may be used to make a small implicit allowance for contingencies  
 There may be statutory rules about whether or not you discount, and what rate should be used.  
 No discounting of liabilities is likely

[2]

iii)

Likelihood of catastrophes and accumulations.  
 Vulnerability of capital to major shocks such as these.  
 Effect of actual non-proportional reinsurance coverages on catastrophes.  
 Spread of different risk groups within the portfolio.  
 Insurer's experience of writing different classes of business.  
 Expected level of profitability, both underwriting profit and level of investment income generated (model tail length of classes).

[3]

iv)

Statutory authorities — to protect policyholders.  
 Company management — duty to company / shareholders / policyholders.  
 Rating agencies — provide accurate information.  
 Brokers — to place business with solvent companies.  
 Shareholders — to monitor their investments.  
 Policyholders — to check the solvency of their insurers.  
 Competitors — to try to discover competitively useful facts.  
 Potential purchasers of the company — to assess its value.

[3]

v)

Might test variability using variations in assumptions with a deterministic model.  
 Easier to test effects of particular scenarios using a deterministic model.  
 Stochastic more complex, and harder to get right.  
 Stochastic harder to parameterise.  
 Stochastic more effective in incorporating allowance for volatility in asset values  
 And uncertainty in claims experience.

Stochastic may take into account correlations between and within assets and liabilities.

Stochastic allows construction of probability distribution for outcomes.

[3]

[15 Marks]

**Solution 8:**

- i) a.) **Back-Testing**: It is the process of comparing actual experience with model output. It tests how well the model predicts the range of outcomes that actually occur. Back testing is essential in order to ensure that the model is a sufficiently accurate reflection of the real world.

[2]

b.) **Reserving Risk**: The risk that the level of technical reserves held is inappropriate to the level and type of business written / earned. The capital impact of the reserving risk is the difference between the eventual cost at the firm's chosen level of risk tolerance of settling claims for the business written / earned before the modelled period, and the current reserves.

[2]

c.) **Reserving Cycle**: There is sometimes a "reserving cycle" in published accounts in that many insurers tend to cut reserves to the bone when the market is soft to avoid having to make results look too adverse, but reserve more strongly in hard markets when they can afford it.

[2]

- ii) Investigate if the carried reserve adequacy has varied significantly over time.

For each evaluation year, look at the initial carried reserve and proportional development of the industry's carried loss from the initial carried reserve to the final paid amount.

OR

For each accident year, look at the initial carried reserve (at twelve months of development) and proportional development of the industry's carried loss from the initial carried reserve to the final paid amount.

Prepare a chart showing ratios of the carried hindsight unpaid loss by evaluation year or accident year at various months of development to the initial carried reserve.

If there was no reserving cycle, these ratios should be reasonably close to 1 and any variations from 1 should be random in nature. And, it must be the case for all lines and not just any one line as reserving cycle is more of systemic nature.

[4]

- iii) The unpaid loss is an estimate of the cost of future payments. Eventually, how much the actual payments cost will be known or the “actual” unpaid – what the reserve should have been, with perfect hindsight. Identify the corresponding percentile on the unpaid distribution.

Perform this exercise for all years under the study and for each year and separately for each insurer and line of business. The focus is only the recent accident year. For example, if the evaluation year is, say, 2000, then consider the carried reserves only for AY 2000 and compare it with the “actual” unpaid for AY 2000.

If the stochastic reserving model gives an accurate indication of the probability of the actual outcome, we should find a uniform distribution of these percentiles. For example, the 90th percentile is a number that the insurer expects to exceed 10% of the time (that is the definition of the 90th percentile). Therefore, we should find 10% of the companies have an actual outcome that falls above the 90th percentile. And similarly, there should be a 10% chance that the actual reserves fall in the 80th to 90th percentile and so on.

This information could also be used in back-testing the assumptions made on correlations between lines of business. It could also be used in back-testing assumptions made on correlations made between accident years.

[6]

[16 Marks]

### Solution 9:

- i) Required Profit =  $10\% \times \text{Capital} = 10\% \times 150 = 15$   
 Premium = Expected Loss + Required Profit =  $122 + 15 = 137$

[2]

- ii)  $E(X_A) = 100$  and  $E(X_B) = 22$

b=287:  $Y > 287$  is feasible only in scenario III.

So  $E[X_A - E(X_A) | Y > b] = (X_A \text{ in scenario III} - 100) = (120 - 100) = 20$

So  $E[X_B - E(X_B) | Y > b] = (X_B \text{ in scenario III} - 22) = (220 - 22) = 198$

(2 marks)

So, allocated capital to line A =  $[20 / (20 + 198)] \times 150 = 13.8 \sim 14$

And, allocated capital to line B =  $150 - 13.76 = 136.2 \sim 136$

b=137:  $Y > 137$  is feasible only in scenarios II and III.

So  $E[X_A - E(X_A) | Y > b] =$

Conditional Probability of Scenario II \*  $(X_A \text{ in scenario II} - 100)$

+ Conditional Probability of Scenario III \*  $(X_A \text{ in scenario III} - 100)$

$= (49\% / (49\% + 1\%)) \times (120 - 100) + (1\% / (49\% + 1\%)) \times (120 - 100) = 20$

$$\begin{aligned}
 &\text{So } E[X_B - E(X_B) | Y > b] = \\
 &\text{Conditional Probability of Scenario II} * (X_B \text{ in scenario II} - 100) \\
 &+ \text{Conditional Probability of Scenario III} * (X_B \text{ in scenario III} - 100) \\
 &= (49\% / (49\% + 1\%)) * (20 - 22) + (1\% / (49\% + 1\%)) * (220 - 22) = -1.96 + 3.96 = 2
 \end{aligned}$$

So, allocated capital to line A =  $[20 / (20 + 2)] * 150 = 136.4 \sim 136$

And, allocated capital to line B =  $150 - 136.4 = 13.6 \sim 14$

[6]

iii) b=287 (1 mark)

|        | Allocated Capital | Required Profit = 10% * Allocated Capital | Premium = Expected Loss + Required Profit |
|--------|-------------------|-------------------------------------------|-------------------------------------------|
| Line A | 14                | 1.4                                       | 101.4                                     |
| Line B | 136               | 13.6                                      | 35.6                                      |

b=137 (1 mark)

|        | Allocated Capital | Required Profit = 10% * Allocated Capital | Premium = Expected Loss + Required Profit |
|--------|-------------------|-------------------------------------------|-------------------------------------------|
| Line A | 136               | 13.6                                      | 113.6                                     |
| Line B | 14                | 1.4                                       | 23.4                                      |

[2]

iv) Value of '287' represents the amount of premium charged for 10% return on capital + available capital. So  $Y > 287$  signifies insolvency.

Here, capital allocated is so much more to line B as line B is posing a disproportionately high risk to insolvency compared to line A.

Value of '137' represents the amount of combined premium to be charged for 10% return on capital. So  $Y > 137$  represents loss making situation.

Here, capital allocated to lines A and B are more in line with their expected losses and solvency risk is not playing a major role as it did for b=287.

As required profits (premiums – expected losses) are 10% of capital, required premiums also change considerably based on the allocated capital.

Generally speaking, since policyholders are concerned only about insolvency and shareholders are concerned about all loss making situations, choice of 287 is more relevant to policyholders and choice of 137 is more relevant to shareholders.

[6]

- v) Required capital is commensurate with risk and therefore more capital may be required for the longer-tailed lines depending on the risk profile.

Capital is held for reserve risk. As relatively more reserves are held for longer-tailed lines, more capital is required.

Capital will be held for a greater time period as reserves are held for longer periods.

Reserves held may be on an undiscounted basis rather than the present value under statutory accounting which could attract some additional cost of capital. For example, if premium is 100 and expected losses are 110 over a 10-year duration with present value of 90, the initial reserves required to be held would be 110 instead of 90. This would need an additional capital of 10.

**[4]**

**[20 Marks]**

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